

# CS23334 Fundamentals of Data Science Lab

## Experiment 13

### Procedure to Handle Missing Values & Inappropriate Data in a Dataset

**Objective:** To demonstrate data preprocessing techniques for identifying duplicates, removing redundant attributes, handling invalid values, standardizing categorical text, and imputing missing data in a hotel booking dataset (`Hotel_Dataset.csv`).

#### Methodology:

- 1 **Duplicate Detection & Removal:** Identify duplicate records using `df.duplicated()` and remove them using `df.drop_duplicates(inplace=True)`, followed by index resetting.
- 2 **Redundant Attribute Removal:** Drop duplicated or unnecessary columns (e.g. `Age_Group.1`) using `df.drop()`.
- 3 **Invalid Data Handling:** Convert negative continuous values (`Bill`, `EstimatedSalary`) and out-of-bounds ratings/pax values to `np.nan`.
- 4 **Categorical Standardization:** Clean up inconsistent text representations (`Ibys` → `Ibis`, `Vegetarian/veg` → `Veg`, `non-Veg` → `Non-Veg`).
- 5 **Statistical Imputation:** Impute missing continuous attributes using feature mean and discrete attributes using feature median.

## Code:

```
import numpy as np
import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")
df

df.duplicated()
df.info()

df.drop_duplicates(inplace=True)
df

len(df)

index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
df

df.drop(['Age_Group.1'], axis=1, inplace=True)
df
```

```

df.CustomerID.loc[df.CustomerID < 0] = np.nan
df.Bill.loc[df.Bill < 0] = np.nan
df.EstimatedSalary.loc[df.EstimatedSalary < 0] = np.nan
df

df['Rating(1-5)'].loc[(df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5)] = np.nan
df['NoOfPax'].loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20)] = np.nan
df

df.Age_Group.unique()
df.Hotel.unique()

df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)

df.FoodPreference.unique()
df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
df.FoodPreference.replace(['non-Veg'], 'Non-Veg', inplace=True)

df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)
df.NoOfPax.fillna(round(df.NoOfPax.median()), inplace=True)
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
df.Bill.fillna(round(df.Bill.mean()), inplace=True)
df

```

## OUTPUT: 1. Initial Dataset, Duplicate Detection & Index Reset

```
In [1]: df = pd.read_csv("Hotel_Dataset.csv")
```

```
In [2]: df.duplicated()
```

```
Out[2]:
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9     True
10   False
dtype: bool
```

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 11 entries, 0 to 10
```

```
Data columns (total 9 columns):
```

| # | Column          | Non-Null Count | Dtype  |
|---|-----------------|----------------|--------|
| 0 | CustomerID      | 11 non-null    | int64  |
| 1 | Age_Group       | 11 non-null    | object |
| 2 | Rating(1-5)     | 11 non-null    | int64  |
| 3 | Hotel           | 11 non-null    | object |
| 4 | FoodPreference  | 11 non-null    | object |
| 5 | Bill            | 11 non-null    | int64  |
| 6 | NoOfPax         | 11 non-null    | int64  |
| 7 | EstimatedSalary | 11 non-null    | int64  |
| 8 | Age_Group.1     | 11 non-null    | object |

```
In [4]: df.drop_duplicates(inplace=True)
```

```
In [5]: index = np.array(list(range(0, len(df))))
```

```
In [6]: df.set_index(index, inplace=True)
```

```
In [7]: df.drop(['Age_Group.1'], axis=1, inplace=True)
```

```
In [8]: df
```

```
Out[8]:
```

|   | CustomerID | Age_Group | Rating(1-5) | Hotel     | FoodPreference | Bill | NoOfPax | EstimatedSalary |
|---|------------|-----------|-------------|-----------|----------------|------|---------|-----------------|
| 0 | 1          | 20-25     | 4           | Ibis      | veg            | 1300 | 2       | 40000           |
| 1 | 2          | 30-35     | 5           | LemonTree | Non-Veg        | 2000 | 3       | 59000           |
| 2 | 3          | 25-30     | 6           | RedFox    | Veg            | 1322 | 2       | 30000           |
| 3 | 4          | 20-25     | -1          | LemonTree | Veg            | 1234 | 2       | 120000          |
| 4 | 5          | 35+       | 3           | Ibis      | Vegetarian     | 989  | 2       | 45000           |
| 5 | 6          | 35+       | 3           | Ibys      | Non-Veg        | 1909 | 2       | 122220          |
| 6 | 7          | 35+       | 4           | RedFox    | Vegetarian     | 1000 | -1      | 21122           |
| 7 | 8          | 20-25     | 7           | LemonTree | Veg            | 2999 | -10     | 345673          |
| 8 | 9          | 25-30     | 2           | Ibis      | Non-Veg        | 3456 | 3       | -99999          |
| 9 | 10         | 30-35     | 5           | RedFox    | non-Veg        | 6755 | 4       | 87777           |

## OUTPUT: 2. Out-of-Bounds Replacement, Categorical Cleaning & Imputed DataFrame

```
In [9]: df.CustomerID.loc[df.CustomerID < 0] = np.nan
In [10]: df.Bill.loc[df.Bill < 0] = np.nan
In [11]: df.EstimatedSalary.loc[df.EstimatedSalary < 0] = np.nan
In [12]: df['Rating(1-5)'].loc[(df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5)] = np.nan
In [13]: df['NoOfPax'].loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20)] = np.nan
```

```
In [14]: df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)
In [15]: df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
In [16]: df.FoodPreference.replace(['non-Veg'], 'Non-Veg', inplace=True)
```

```
In [17]: df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)
In [18]: df.NoOfPax.fillna(round(df.NoOfPax.median()), inplace=True)
In [19]: df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
In [20]: df.Bill.fillna(round(df.Bill.mean()), inplace=True)
In [21]: df
```

Out[21]:

|   | CustomerID | Age_Group | Rating(1-5) | Hotel     | FoodPreference | Bill   | NoOfPax | EstimatedSalary |
|---|------------|-----------|-------------|-----------|----------------|--------|---------|-----------------|
| 0 | 1.0        | 20-25     | 4.0         | Ibis      | Veg            | 1300.0 | 2.0     | 40000.0         |
| 1 | 2.0        | 30-35     | 5.0         | LemonTree | Non-Veg        | 2000.0 | 3.0     | 59000.0         |
| 2 | 3.0        | 25-30     | 4.0         | RedFox    | Veg            | 1322.0 | 2.0     | 30000.0         |
| 3 | 4.0        | 20-25     | 4.0         | LemonTree | Veg            | 1234.0 | 2.0     | 120000.0        |
| 4 | 5.0        | 35+       | 3.0         | Ibis      | Veg            | 989.0  | 2.0     | 45000.0         |
| 5 | 6.0        | 35+       | 3.0         | Ibis      | Non-Veg        | 1909.0 | 2.0     | 122220.0        |
| 6 | 7.0        | 35+       | 4.0         | RedFox    | Veg            | 1000.0 | 2.0     | 21122.0         |
| 7 | 8.0        | 20-25     | 4.0         | LemonTree | Veg            | 2999.0 | 2.0     | 345673.0        |
| 8 | 9.0        | 25-30     | 2.0         | Ibis      | Non-Veg        | 3456.0 | 3.0     | 96755.0         |
| 9 | 10.0       | 30-35     | 5.0         | RedFox    | Non-Veg        | 6755.0 | 4.0     | 87777.0         |